

# Cross-Survey Consistency of BAO under Flat $\Lambda$ CDM: SDSS/eBOSS DR16 versus DESI DR2, Fitted Separately, with a Kernel-Verified Tension Statistic

AutoevolveAI / ANSE Research Pipeline

Model-produced by an automated Claude Code pipeline

(the generating model version is not recorded in the result artifacts);

reviewed only by an automated model referee, not by a person

September 27, 2026

## Abstract

We fit flat- $\Lambda$ CDM, BAO-only, to two surveys *separately*: the completed SDSS/eBOSS DR16 BAO compilation [1], using its distributed grid likelihoods, and DESI DR2 BAO [2]. We then compare the two posteriors in the  $(\Omega_m, hr_d)$  plane. SDSS gives  $\Omega_m = 0.2987 \pm 0.0164$  and  $hr_d = (100.52 \pm 1.29)$  Mpc. DESI DR2 gives  $\Omega_m = 0.2980 \pm 0.0087$  and  $hr_d = (101.51 \pm 0.75)$  Mpc. These reproduce the published values to within  $|\text{pull}| \leq 0.06\sigma$  on the three targets we could check against primary sources.

The parameter-difference statistic  $\chi^2 = \Delta p^\top (C_{\text{SDSS}} + C_{\text{DESI}})^{-1} \Delta p$  gives  $\chi^2 = 1.932$  for 2 degrees of freedom, a probability to exceed (PTE) of 0.381, and  $N_\sigma = 0.877$ . This passes the preregistered criterion  $N_\sigma < 2$  and agrees with DESI DR2’s qualitative statement that its BAO are consistent with SDSS. The statistic treats the two surveys as independent. Their volumes overlap; if the cross-survey covariance  $K$  has  $K + K^\top$  positive semidefinite, this is a lower bound with respect to that overlap ( $K$  was not estimated).

The analysis choices, pass criteria and controls were preregistered before any fit. An injected-parameter positive control, run as preregistered on the SDSS Gaussian-summary variant and on DESI (not on the SDSS baseline grid likelihood), and the gated negative controls behave as required. In Lean 4 we kernel-verify nine algebraic properties of the tension statistic, including that it equals Mathlib’s  $d \cdot (C^{-1}d)$ , is symmetric in the two surveys, and is a positive-definite quadratic form. All nine report exactly the trusted axioms. This is a reproduction and consistency exercise. It is not a new cosmological constraint.

## 1 Introduction

BAO surveys measure the transverse comoving distance  $D_M(z)/r_d$ , the Hubble distance  $D_H(z)/r_d = c/(H(z)r_d)$ , and the isotropic combination  $D_V(z)/r_d = [zD_M^2 D_H]^{1/3}/r_d$ . In flat  $\Lambda$ CDM,  $E(z) = \sqrt{\Omega_m(1+z)^3 + 1 - \Omega_m}$ , and BAO-only data constrain exactly two parameters,  $\Omega_m$  and  $hr_d$ . DESI DR2 states that it samples these two parameters when fitting BAO alone [2].

DESI DR2 says its BAO results are “consistent with DESI DR1 and SDSS”. In a four-bin comparison it concludes “there is no significant discrepancy” (largest differences  $1.49\sigma$  in  $\alpha_{\text{iso}}$  and  $1.69\sigma$  in  $\alpha_{\text{AP}}$ ), and it gives the LRG2  $D_M/r_d$  difference against eBOSS as  $1.5\sigma$  to  $2.6\sigma$  [2]. It publishes no tension number for SDSS versus DR2 in the  $(\Omega_m, hr_d)$  plane. The DR1 analysis defines a parameter-difference statistic, its eq. (4.2), which it applies to the comparison of DESI DR1 BAO with the CMB (§4.1); its comparison of DR1 with SDSS (§3.3) is qualitative only, reporting “no

significant difference in  $\Omega_m$  and a shift of just  $\sim 1\sigma$  in  $r_d h$ ” [3]. This paper computes the eq. (4.2) number for SDSS versus DR2 from the public data products, fitting the two surveys separately and never combining them.

The exercise follows an earlier flat- $\Lambda$ CDM reproduction of DESI DR1 in the same repository (`papers/bao_flatcdm/`), which matched DESI’s published values to within  $0.07\sigma$  and  $0.11\sigma$ . It adds three things: non-Gaussian grid likelihoods, a preregistration fixed before any fit, and a formalization of the comparison statistic itself.

## 2 Data

All input files come from a local copy of a BAO data collection. Its README lists the DESI DR2 and DR1 files, and the eBOSS DR16, SDSS DR7 MGS and SDSS DR12 files “as originally distributed with CosmoMC”. The likelihood conventions (redshifts, table formats, the MGS  $\alpha$  rescaling) were read from the Cobaya source [5]. Each file’s SHA-256 was fixed in the preregistration and re-verified by the fit (`input_sha256_verified: true`).

**SDSS/eBOSS DR16 [1].** The baseline likelihood is the sum of the pieces below, with no covariance between samples, as in [1]:

- MGS ( $z = 0.15$ ): a  $\chi^2(\alpha)$  table, with  $\alpha = (D_V/r_d)/4.29720761315$ .
- Gaussian pieces (Table 1): BOSS DR12 galaxies at  $z = 0.38$  and  $0.51$ , eBOSS LRG at  $z = 0.698$ , and eBOSS QSO at  $z = 1.48$ , with the distributed covariances.
- eBOSS ELG ( $z = 0.845$ ): a 1D likelihood table in  $D_V/r_d$ .
- $\text{Ly}\alpha$  auto and  $\text{Ly}\alpha \times \text{QSO}$  ( $z = 2.334$ ): two 2D likelihood grids in  $(D_M/r_d, D_H/r_d)$ , interpolated in  $\ln L$  with bicubic splines and an explicit bounds check.

Three sets of files were excluded before any fit:

- the DESI DR1 + eBOSS combined  $\text{Ly}\alpha$  file, because it contains DESI data. The SDSS likelihood reads only a fixed dictionary of SDSS filenames whose SHA-256 hashes are preregistered; a code guard checks the filename, the allow-list and the hash, and raises otherwise (Section 5);
- the DR12 consensus file, whose  $z = 0.61$  bin double-counts the eBOSS LRG volume;
- all files that include RSD or full-shape information.

A *Gaussian-summary* variant serves as a sensitivity check and as the basis of the positive control. It replaces MGS and ELG with the Table III values of [1], and replaces the two  $\text{Ly}\alpha$  grids with the joint auto+cross result of [4], eq. (43).

**DESI DR2 [2].** The data are `desi_gaussian_bao_ALL_GCcomb_{mean,cov}.txt`: 13 data points with a  $13 \times 13$  covariance (Table 2).

Table 1: Gaussian pieces of the SDSS likelihood, read from the distributed files.  $\sigma = \sqrt{\text{diag } C}$ . The DR12 file gives  $\sigma(D_H, 0.38) = 0.729$ , while [1] Table III rounds it to 0.76. The file is used, and this difference was recorded before the fit.

| Sample    | $z$   | Quantity  | Value  | $\sigma$ |
|-----------|-------|-----------|--------|----------|
| BOSS DR12 | 0.380 | $D_M/r_d$ | 10.234 | 0.169    |
| BOSS DR12 | 0.380 | $D_H/r_d$ | 24.981 | 0.729    |
| BOSS DR12 | 0.510 | $D_M/r_d$ | 13.366 | 0.204    |
| BOSS DR12 | 0.510 | $D_H/r_d$ | 22.317 | 0.572    |
| eBOSS LRG | 0.698 | $D_M/r_d$ | 17.858 | 0.328    |
| eBOSS LRG | 0.698 | $D_H/r_d$ | 19.326 | 0.533    |
| eBOSS QSO | 1.480 | $D_M/r_d$ | 30.688 | 0.798    |
| eBOSS QSO | 1.480 | $D_H/r_d$ | 13.261 | 0.552    |

Table 2: DESI DR2 BAO data vector as distributed.  $\sigma = \sqrt{\text{diag } C}$ . The fit uses the full covariance.

| Survey   | $z$   | Quantity  | Value  | $\sigma$ |
|----------|-------|-----------|--------|----------|
| DESI DR2 | 0.295 | $D_V/r_d$ | 7.942  | 0.076    |
| DESI DR2 | 0.510 | $D_M/r_d$ | 13.588 | 0.168    |
| DESI DR2 | 0.510 | $D_H/r_d$ | 21.863 | 0.429    |
| DESI DR2 | 0.706 | $D_M/r_d$ | 17.351 | 0.180    |
| DESI DR2 | 0.706 | $D_H/r_d$ | 19.455 | 0.334    |
| DESI DR2 | 0.934 | $D_M/r_d$ | 21.576 | 0.162    |
| DESI DR2 | 0.934 | $D_H/r_d$ | 17.641 | 0.201    |
| DESI DR2 | 1.321 | $D_M/r_d$ | 27.601 | 0.325    |
| DESI DR2 | 1.321 | $D_H/r_d$ | 14.176 | 0.225    |
| DESI DR2 | 1.484 | $D_M/r_d$ | 30.512 | 0.764    |
| DESI DR2 | 1.484 | $D_H/r_d$ | 12.817 | 0.518    |
| DESI DR2 | 2.330 | $D_H/r_d$ | 8.632  | 0.101    |
| DESI DR2 | 2.330 | $D_M/r_d$ | 38.989 | 0.532    |

### 3 Method

The predictions are  $D_M/r_d = (c/100 \text{ hr}_d) \int_0^z dz'/E(z')$ ,  $D_H/r_d = c/(100 \text{ hr}_d E(z))$  and the  $D_V$  combination above. Radiation is neglected. Section 5 checks this against CAMB. Three prediction codes are required to agree to  $10^{-4}$  relative: `astropy FlatLambdaCDM`, `scipy.integrate.quad`, and Gauss–Legendre quadrature. Their measured maximum disagreement was  $9.3 \times 10^{-15}$ .

The priors are flat, with  $\Omega_m \in [0.05, 0.95]$  and  $\text{hr}_d \in [60, 150]$  Mpc. Each dataset is sampled with `emcee` (32 walkers, 6000 steps, burn-in 1000). Every chain satisfies  $50 \tau < 6000$ , with a maximum autocorrelation time  $\tau$  of 30.6 for DESI and 29.0 for SDSS. We report the posterior mean  $\pm$  standard deviation, which is the DESI convention.

There are two independent cross-checks:

- a normalized  $301 \times 301$  grid posterior; its means agree with `emcee` within  $0.014\sigma$  for SDSS and  $0.020\sigma$  for DESI;
- a MAP estimate with Laplace errors.

**Tension statistic.** With  $p = (\Omega_m, hr_d)$ , the statistic is  $\chi^2 = \Delta p^\top (C_{\text{SDSS}} + C_{\text{DESI}})^{-1} \Delta p$ . This is eq. (18) of [2] and eq. (4.2) of [3]. It is converted to a PTE for 2 degrees of freedom and then to a two-sided Gaussian  $N_\sigma$ . The code is `baolib.tension`. We also report the 1D parameter differences and a KDE parameter-shift PTE on the chains, which does not assume Gaussian posteriors.

## 4 Preregistration

`results/eboss_vs_desi/preregistration.json` (SHA-256 prefix `e0eca89f279cfaac`) was written at 2026-09-27T15:50Z and amended once, before any fit. The fit output records that hash, and it was unchanged at the time of the fit. The preregistration fixed the following:

- **Targets.**
  - SDSS  $\Omega_m = 0.299 \pm 0.016$ , from Table IV of [1]:  $\Omega_{\text{DE}} = 0.701^{+0.017}_{-0.015}$ .
  - DESI DR2  $\Omega_m = 0.2975 \pm 0.0086$  and  $hr_d = (101.54 \pm 0.73)$  Mpc, from eq. (17) of [2].
  - SDSS  $hr_d = (100.4 \pm 1.3)$  Mpc. This is quoted by [3] §4.1, which cites [1], but it was not found in the fetched pages of [1]. It is therefore reported but not counted.
- **Tolerance.**  $|\text{pull}| < 0.5\sigma$  on the three primary-verified targets.
- **Consistency criterion.**  $N_\sigma < 2$ .
- **Expected range.** From the published means alone,  $N_\sigma$  falls between 0.32 and 1.56 as the unpublished SDSS  $\Omega_m$ – $hr_d$  correlation is scanned over  $[-0.95, 0.95]$ . This range is an expectation, not a criterion.
- **Controls.** All controls and their thresholds (Section 5).

## 5 Controls

**Instrument check on the grid likelihoods** (no cosmology). Each interpolated table must reproduce its published peak and width:

- MGS: peak 4.466 versus 4.466, which is  $0.0003\sigma$  off; width ratio 0.941.
- ELG: peak 18.328 versus 18.330; width ratio 1.007. The distributed table is truncated at its lower edge,  $D_V/r_d = 14.88$ , where  $L/L_{\text{max}}$  is still 0.19; below that edge our likelihood returns  $-\infty$  in  $\ln L$ . This does not affect the real fit: the ELG term contributes 0.27 to  $-2 \ln L$  at the SDSS MAP.
- Ly $\alpha$  auto: peak  $(D_M, D_H) = (37.60, 8.930)$  versus  $(37.60, 8.930)$ ;  $\rho = -0.476$  versus  $-0.49$ ; width ratios 0.91/0.97.
- Ly $\alpha$ ×QSO: peak  $(37.33, 9.082)$  versus  $(37.30, 9.080)$ ;  $\rho = -0.441$  versus  $-0.43$ ; width ratios 0.97/0.84. The preregistered width tolerance for Ly $\alpha$  is 0.25.
- Paired negative: dropping the MGS  $\alpha$  rescaling must fail. It does, with a peak at 1.039,  $20.4\sigma$  off.

**Positive control** (passed). There were 200 realizations of synthetic Gaussian data, injected at (0.2975, 101.54) with seed 20260927. The preregistered thresholds are  $1\sigma$  coverage in  $[0.60, 0.76]$  and  $|\text{bias}| < 0.2\sigma$ .

- SDSS Gaussian-summary: coverage 0.695 for  $\Omega_m$  and 0.675 for  $hr_d$ ; bias  $0.045\sigma$  and  $0.077\sigma$ .
- DESI DR2: coverage 0.655 and 0.665; bias  $0.066\sigma$  and  $0.086\sigma$ .
- Tension calibration on pairs drawn from the same parameters: median  $N_\sigma = 0.672$ ; KS test of the PTE against a uniform distribution,  $p = 0.549$ ; fraction with  $N_\sigma > 2$  equal to 0.030.
- emcee subset of 20 realizations: coverage 0.75/0.75 for SDSS and 0.65/0.60 for DESI.
- Not gated, reported for completeness: a KS test of the injected realizations'  $\chi^2_{\min}$  against  $\chi^2(\text{dof})$  gives  $p = 0.085$  (SDSS Gaussian-summary) and  $p = 0.073$  (DESI DR2), so the fit-quality distribution is only marginally consistent with the nominal one.

This control never exercises the SDSS baseline grid likelihood (MGS, ELG and Ly $\alpha$  tables). It runs on the Gaussian-summary variant only, as preregistered. For the baseline, the evidence is indirect: the instrument check below reproduces every table's published peak and width, and the baseline and variant posteriors differ by  $0.032\sigma$  in  $\Omega_m$  and  $0.004\sigma$  in  $hr_d$ .

**Negative controls** (all rejected).

- No- $\Lambda$  model (Einstein–de Sitter) on DESI:  $\Delta(-2\ln L) = 1409.9$ , against a required  $> 25$ . On the SDSS baseline the value is  $\infty$ . The preregistered boolean passes, but only *by construction*:  $\Omega_m = 1$  predictions fall outside the grid tables for every  $hr_d$ , so the test does not discriminate. A probe likelihood that returns  $+\infty$  everywhere passes the as-coded boolean too (as-coded boolean: rejected). A stricter gate was therefore added on top of the preregistered one (recorded as such in `negative_controls.json`): it also requires both minima to be finite, and the SDSS EdS test is gated on the Gaussian-summary variant,  $\Delta = 376.4$ . Under the stricter gate the SDSS baseline is recorded as non-discriminating and the  $+\infty$  probe is correctly not rejected (probe check passed).
- DESI  $D_M \leftrightarrow D_H$  swap:  $\chi^2/\text{dof} = 5248.7$ , against a required  $> 3$ .
- DESI covariance divided by 25:  $\chi^2/\text{dof} = 23.34$ , against a required  $> 3$ .
- Every DESI ratio multiplied by 0.93:  $N_\sigma = 12.14$  against SDSS. The requirement was  $> 3$ , and the closed-form expectation was at least 5.44.
- Combination guard (passed): it raised on the excluded DESI+eBOSS Ly $\alpha$  file (filename check), on a copy of the DESI DR2 mean file under a non-DESI name (allow-list check), and on the same copy renamed to a real SDSS filename (SHA-256 check); all ten real SDSS files were accepted. The guard is a safeguard against wiring mistakes, not a proof that no DESI information enters the SDSS files.
- Report-only permutation of the SDSS covariance:  $-2\ln L_{\min} = 13.51$ , versus 11.52 unpermuted.

**CAMB background cross-check** (passed). CAMB 2.0.4 was run at three parameter points. The largest relative differences between the analytic model and CAMB at the data redshifts are  $2.0 \times 10^{-4}$  in  $D_M$ ,  $4.8 \times 10^{-4}$  in  $H$  and  $2.9 \times 10^{-4}$  in  $D_V$ , all below the  $10^{-3}$  gate. At the Planck-2018 point, CAMB gives  $r_{\text{drag}} = 147.1027$  Mpc.

## 6 Results

Table 3: Flat- $\Lambda$ CDM BAO-only posteriors, each survey fitted separately. The primary estimate is the emcee mean  $\pm$  standard deviation. Pulls are measured against the published values and use the published  $\sigma$ . †: secondary-sourced target, not counted toward the tolerance.

| Dataset                           | $\Omega_m$          | $hr_d$ [Mpc]            | corr     | $-2 \ln L_{\min}$ (dof) |
|-----------------------------------|---------------------|-------------------------|----------|-------------------------|
| SDSS/eBOSS DR16 (baseline, emcee) | $0.2987 \pm 0.0164$ | $100.52 \pm 1.29$       | $-0.880$ | 11.52 (12, nominal)     |
| grid                              | $0.2987 \pm 0.0162$ | $100.53 \pm 1.27$       |          |                         |
| MAP (Laplace $\sigma$ )           | $0.2974 \pm 0.0158$ | $100.58 \pm 1.24$       |          |                         |
| SDSS Gaussian-summary variant     | $0.2993 \pm 0.0165$ | $100.52 \pm 1.26$       | $-0.876$ | 10.87 (10)              |
| DESI DR2 (emcee)                  | $0.2980 \pm 0.0087$ | $101.51 \pm 0.75$       | $-0.926$ | 10.27 (11)              |
| grid                              | $0.2978 \pm 0.0086$ | $101.52 \pm 0.74$       |          |                         |
| MAP (Laplace $\sigma$ )           | $0.2975 \pm 0.0086$ | $101.54 \pm 0.74$       |          |                         |
| Published SDSS [1, 3]             | $0.299 \pm 0.016$   | $100.4 \pm 1.3^\dagger$ |          |                         |
| Published DESI DR2 [2]            | $0.2975 \pm 0.0086$ | $101.54 \pm 0.73$       | $-0.92$  | 10.2 (11)               |
| Pull SDSS                         | $-0.017\sigma$      | $+0.090\sigma^\dagger$  |          |                         |
| Pull DESI DR2                     | $+0.055\sigma$      | $-0.043\sigma$          |          |                         |

Table 4: SDSS versus DESI DR2 parameter-difference statistic (2 dof, surveys treated as independent).

| Estimator                           | $\chi^2$ | PTE   | $N_\sigma$   | 1D $\Omega_m$ | 1D $hr_d$     |
|-------------------------------------|----------|-------|--------------|---------------|---------------|
| emcee moments (primary)             | 1.932    | 0.381 | <b>0.877</b> | $0.041\sigma$ | $0.668\sigma$ |
| grid moments                        | 1.889    | 0.389 | 0.862        | $0.045\sigma$ | $0.675\sigma$ |
| MAP + Laplace                       | 2.092    | 0.351 | 0.932        | $0.004\sigma$ | $0.668\sigma$ |
| Gaussian-summary SDSS               | 1.759    | 0.415 | 0.815        | $0.068\sigma$ | $0.673\sigma$ |
| KDE parameter shift (20000 samples) | —        | 0.348 | 0.939        | —             | —             |

All three primary-verified targets are reproduced within tolerance (`within_tolerance = true`). The SDSS baseline and the Gaussian-summary variant differ by only  $0.032\sigma$  in  $\Omega_m$  and  $0.004\sigma$  in  $hr_d$ . The non-Gaussian grid likelihoods therefore do not drive the result.

The five tension estimators range from 0.815 to  $0.939\sigma$ . The primary value,  $N_\sigma = 0.877$ , satisfies the preregistered  $N_\sigma < 2$ . Most of the difference lies in  $hr_d$  ( $0.668\sigma$  in 1D). The  $\Omega_m$  values agree closely ( $0.041\sigma$ ). The measured SDSS correlation,  $-0.880$ , puts the result inside the preregistered expected range, between the values for correlations  $-0.8$  ( $0.74\sigma$ ) and  $-0.9$  ( $1.10\sigma$ ).

## 7 Lean 4 formalization of the tension statistic

`formal/ANSE/BAO_Consistency.lean` (SHA-256 prefix `560e3b5d8266a2e2`, namespace `ANSE.BAOConsistency`) formalizes the statistic that `bao_lib.tension` evaluates: `dp = p1 -`

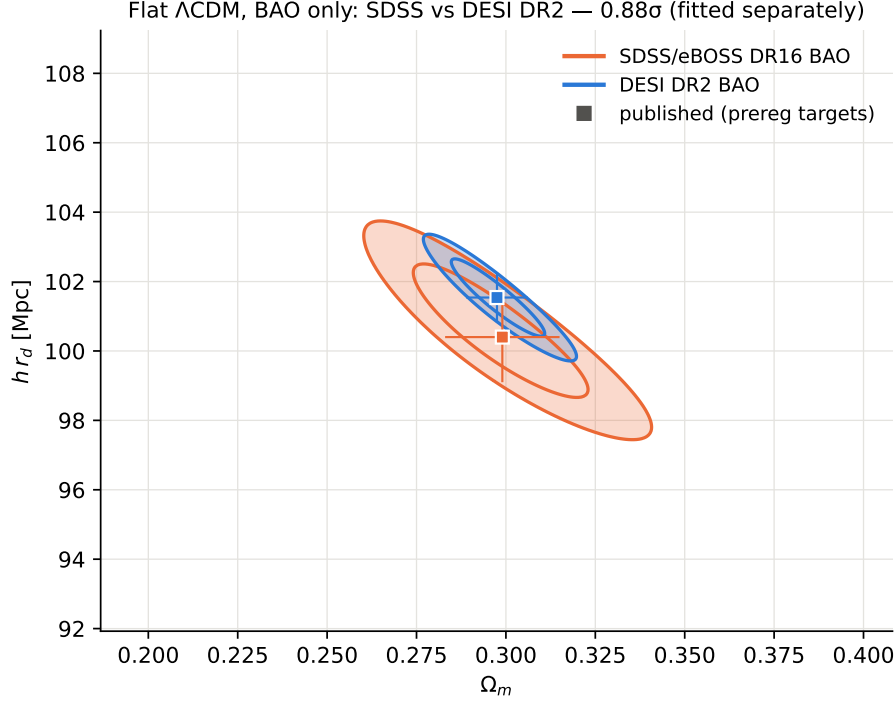


Figure 1: 68.3% and 95.4% highest-density contours of the normalized grid posteriors, with SDSS/e-BOSS DR16 in orange and DESI DR2 in blue. The squares with error bars are the preregistered published targets.

`p2, C = c1 + c2, chi2 = dp @ solve(C, dp)`. The definitions are:

$$\text{tension1D } \mu_1 \sigma_1 \mu_2 \sigma_2 = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2},$$

$$\text{chi2 } a b c d_1 d_2 = \frac{cd_1^2 - 2bd_1d_2 + ad_2^2}{ac - b^2},$$

and `surveyTension` is `chi2` applied to  $C_S + C_D$  and  $p_S - p_D$ . The imports are pinned to the locally built Mathlib subset, with no `import Mathlib`. The toolchain is `leanprover/lean4:v4.34.0-rc2`.

The theorem statements are extracted verbatim from the source by `make_paper_inputs.py`. Proofs are elided, and the Unicode glyphs are typeset as math symbols ( $\cdot_v$  is Mathlib's `dotProduct` and  $\ast_v$  is `mulVec`):

```
theorem tension1D_symm (μ1 σ1 μ2 σ2 : ℝ) :
  tension1D μ1 σ1 μ2 σ2 = tension1D μ2 σ2 μ1 σ1 := ...

theorem tension1D_nonneg (μ1 σ1 μ2 σ2 : ℝ) : 0 ≤ tension1D μ1 σ1 μ2 σ2 := ...

theorem tension1D_eq_zero_iff {μ1 σ1 μ2 σ2 : ℝ} (hσ : 0 < σ1) :
  tension1D μ1 σ1 μ2 σ2 = 0 ↔ μ1 = μ2 := ...

theorem chi2_eq_dotProduct_inv_mulVec (a b c d1 d2 : ℝ) (hdet : a * c - b ^ 2 ≠ 0) :
  ![d1, d2] ·v ((!![a, b; b, c])-1 ·v ![d1, d2]) = chi2 a b c d1 d2 := ...
```

```

theorem chi2_nonneg {a b c : ℝ} (ha : 0 < a) (hdet : 0 < a * c - b ^ 2) (d1 d2 : ℝ) :
  0 ≤ chi2 a b c d1 d2 := ...

theorem chi2_eq_zero_iff {a b c : ℝ} (ha : 0 < a) (hdet : 0 < a * c - b ^ 2) (d1 d2 : ℝ)
  :
  chi2 a b c d1 d2 = 0 ↔ d1 = 0 ∧ d2 = 0 := ...

theorem surveyTension_symm (mS1 mS2 aS bS cS mD1 mD2 aD bD cD : ℝ) :
  surveyTension mS1 mS2 aS bS cS mD1 mD2 aD bD cD
  = surveyTension mD1 mD2 aD bD cD mS1 mS2 aS bS cS := ...

theorem posdef_add {aS bS cS aD bD cD : ℝ}
  (haS : 0 < aS) (hdS : 0 < aS * cS - bS ^ 2)
  (haD : 0 < aD) (hdD : 0 < aD * cD - bD ^ 2) :
  0 < aS + aD ∧ 0 < (aS + aD) * (cS + cD) - (bS + bD) ^ 2 := ...

theorem surveyTension_nonneg_and_zero_iff {mS1 mS2 aS bS cS mD1 mD2 aD bD cD : ℝ}
  (haS : 0 < aS) (hdS : 0 < aS * cS - bS ^ 2)
  (haD : 0 < aD) (hdD : 0 < aD * cD - bD ^ 2) :
  0 ≤ surveyTension mS1 mS2 aS bS cS mD1 mD2 aD bD cD ∧
  (surveyTension mS1 mS2 aS bS cS mD1 mD2 aD bD cD = 0 ↔ mS1 = mD1 ∧ mS2 = mD2) :=
  ...

```

The file was compiled with `lake env lean` in the round-2 fix stage of this pipeline, after its last edit (a docstring rewording on `tension1D_eq_zero_iff`; source sha256 prefix 560e3b5d8266a2e2), and exited with code 0, with no errors, no warnings and no `sorry`. The file's own `#print axioms` lines produced the output below. It is verbatim except that each line is wrapped after the theorem name, and the final `rc=0` line is the exit code, appended by the shell:

```

'ANSE.BAOConsistency.tension1D_symm'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOConsistency.tension1D_nonneg'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOConsistency.tension1D_eq_zero_iff'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOConsistency.chi2_eq_dotProduct_inv_mulVec'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOConsistency.chi2_nonneg'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOConsistency.chi2_eq_zero_iff'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOConsistency.surveyTension_symm'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOConsistency.posdef_add'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOConsistency.surveyTension_nonneg_and_zero_iff'
  depends on axioms: [propext, Classical.choice, Quot.sound]
rc=0

```

Every footprint is exactly the trusted set `{propext, Classical.choice, Quot.sound}`. The SOCRATEAI-SCIENTIFIC-ELENCHUS scanner (`elenchus_check.py`) reported “no findings”.

**What the Lean results do and do not say.** The Lean results are exact facts about the statistic:

- the closed-form `chi2` equals Mathlib’s matrix expression, so it is not an ad hoc formula;
- the statistic is symmetric under swapping SDSS and DESI;
- it is non-negative, and zero exactly when the two means coincide, whenever both posterior covariances are positive definite;
- the matrix inverted by the fit is positive definite whenever both inputs are (`posdef_add`).

The file asserts nothing numeric. The values  $\chi^2 = 1.932$  and  $N_\sigma = 0.877$  are not kernel-checked, and neither is the conversion from PTE to  $N_\sigma$ . The match between the Lean definitions and the Python code is a reading of both, not a kernel fact. In the claim ledger it is filed at Tier C (Section 8).

## 8 Claim ledger

The claims are registered in Elenchus’s tier-capped ledger format in `results/eboss_vs_desi/ledger/`. There are 27 claims, each with a content-addressed evidence blob:

- 9 at Tier A: the Lean axiom footprints (statement audit by an automated model referee, not a person);
- 8 at Tier B: the harness outputs;
- 8 at Tier L: the citations, and every comparison with a published number;
- 2 at Tier C: the lower-bound argument, and the claim that the Lean definitions match the code.

The gate (`tools/ledger.py --evidence-dir`) reports 0 block findings and 0 flags, and exits with code 0.

That exit code needs a caveat about who did the audit. Before the fix round, every Tier A row had `audit: null`. The gate then raised one `LEDGER_UNAUDITED_TIER_A` flag per row and exited with code 1. In the round-2 fix stage, the Lean statement audit of the round-2 referee (`results/eboss_vs_desi/referee_report_round2.json`) was recorded on the ledger rows. That referee judged all nine statements faithful to the code. *It is an automated model referee, not a person.* Each audit object says so (`auditor_is_person: false, human_audit: "open"`). Elenchus’s own design treats this audit as a person’s act. Exit code 0 therefore means that a model referee judged the statements faithful. No human statement audit has happened yet.

The referee audited the source before the docstring rewording it asked for. The ledger builder checks that the saved diff between the two versions touches no declaration or proof line before it carries the audit over.

Four hand-built mutations of the ledger are each caught. Three produce a block finding: refiling a comparison-with-published claim at Tier B (`LEDGER_TIER_INVERSION`), changing one byte of an evidence blob (`LEDGER_EVIDENCE_MISMATCH`), and refiling a citation at Tier A (`LEDGER_KIND_OVERCLAIM`). The fourth, removing one Tier A audit object, brings back the `LEDGER_UNAUDITED_TIER_A` flag and exit 1. The gate therefore discriminates on this ledger’s content, and the exit 0 depends on the recorded audits.

## 9 Deviations from the preregistration

- The planned Lean module `BAO_TensionStatistic.lean`, which covered only the diagonal case, was delivered as `BAO_Consistency.lean`. That file covers the full  $2 \times 2$  covariance and adds the link to Mathlib’s matrix inverse.
- The planned outputs `fit_results.json` and `controls.json` became `fit.json` plus one JSON file per control stage.
- As preregistered, the injection test uses the SDSS Gaussian-summary variant, not the baseline grid likelihood. For the baseline, the evidence is the instrument check and the  $0.032\sigma/0.004\sigma$  shift between the baseline and the variant.
- Provenance flag: `preregistration.json` was last modified on disk at 15:54Z, but its field `timestamp_last_amended_utc` reads 15:58Z, so that field is not the actual write time. The frozen file is not edited; the file-system time (15:54Z) is recorded as the amendment time in the ledger README. Both times precede the first control output (15:59Z, from an interrupted earlier attempt), and `prereg_inputs.json` (15:53Z) already contains the literature-only computation that motivated amendment (a). Every output was later regenerated.
- Gates added beyond the preregistration (all tighten, none loosen): the Einstein–de Sitter gate also requires finite minima and uses the SDSS Gaussian-summary variant for SDSS; the combination guard checks an allow-list and the preregistered hashes in addition to the filename, with two renamed-file probes; non-finite values in `negative_controls.json` are written as strings. All control and fit outputs used here were regenerated by the numerics stage.

## 10 Limitations

- **Survey overlap.** The statistic assumes the surveys are independent. DESI DR2 estimates a correlation  $C \approx 0.57$  between its LRG2 sample and the eBOSS LRGs [2], and the footprints overlap. If the cross-covariance  $K$  makes  $K + K^\top$  positive semidefinite, the correct matrix  $C_S + C_D - K - K^\top$  can only increase  $\chi^2$ .  $N_\sigma = 0.877$  is therefore a lower bound with respect to overlap. We did not estimate  $K$ .
- **Grid-likelihood degrees of freedom.** For the SDSS baseline, the degrees of freedom are nominal, so  $-2 \ln L_{\min}$  should not be read as a  $\chi^2$  goodness of fit.
- **Secondary target.** The SDSS  $hr_d$  target is secondary-sourced.
- **Different pipelines.** Our priors and parameterization differ from eBOSS’s CosmoMC setup, which sampled  $(\Omega_m, H_0, \Omega_b)$ .
- **Model simplifications.** Radiation is neglected (a sub- $10^{-3}$  effect; see the CAMB check), and massive neutrinos are absorbed into  $\Omega_m$ .
- **Repository checks.** `antigravity_guard.py` fails. In the numerics stage its only violation under this exercise was `import camb` in `camb_crosscheck.py` (CAMB is installed only in a separate environment), which was left failing rather than worked around. Re-run with system `python3` in the ledger/paper stage and again in the round-2 fix stage, it reports 74 repository-wide unresolved-import violations, none of them in `scripts/eboss_vs_desi/`; the guard’s result depends on which interpreter runs it. `test_rigor_guard.py` passed. The final pipeline

did not run the repository suite `pytest tests/`. An earlier, interrupted fix round did run it. Its log (`results/eboss_vs_desi/pytest_tests.log`, not regenerated) reports 22 failed, 1358 passed, 49 skipped and 8 errors. None of the failing or erroring tests concerns this exercise: the log contains no line mentioning `eboss_vs_desi`. The CLAUDE.md completion criterion (all three repository checks pass) is therefore *not* met.

- **Import wiring.** `formal/ANSE.lean` does not yet import the new Lean module.

## References

- [1] Alam et al., “Completed SDSS-IV eBOSS: cosmological implications from two decades of spectroscopic surveys,” *Phys. Rev. D* **103**, 083533, arXiv:2007.08991.
- [2] DESI Collaboration, “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints,” arXiv:2503.14738 (2025).
- [3] DESI Collaboration, “DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations,” arXiv:2404.03002 (2024).
- [4] du Mas des Bourboux et al., eBOSS DR16  $\text{Ly}\alpha$  BAO, arXiv:2007.08995.
- [5] Cobaya BAO likelihood source and data conventions, <https://github.com/CobayaSampler/cobaya> (files `cobaya/likelihoods/bao/*.yaml` and `cobaya/likelihoods/base_classes/bao.py`), read 2026-09-27.